THOMAS MORE

Integrated Lab

Data case
Bilal Ozden

THOMAS MORE

Special Olympics

Special Olympics

Data Pipeline & Analytics

1. Resources

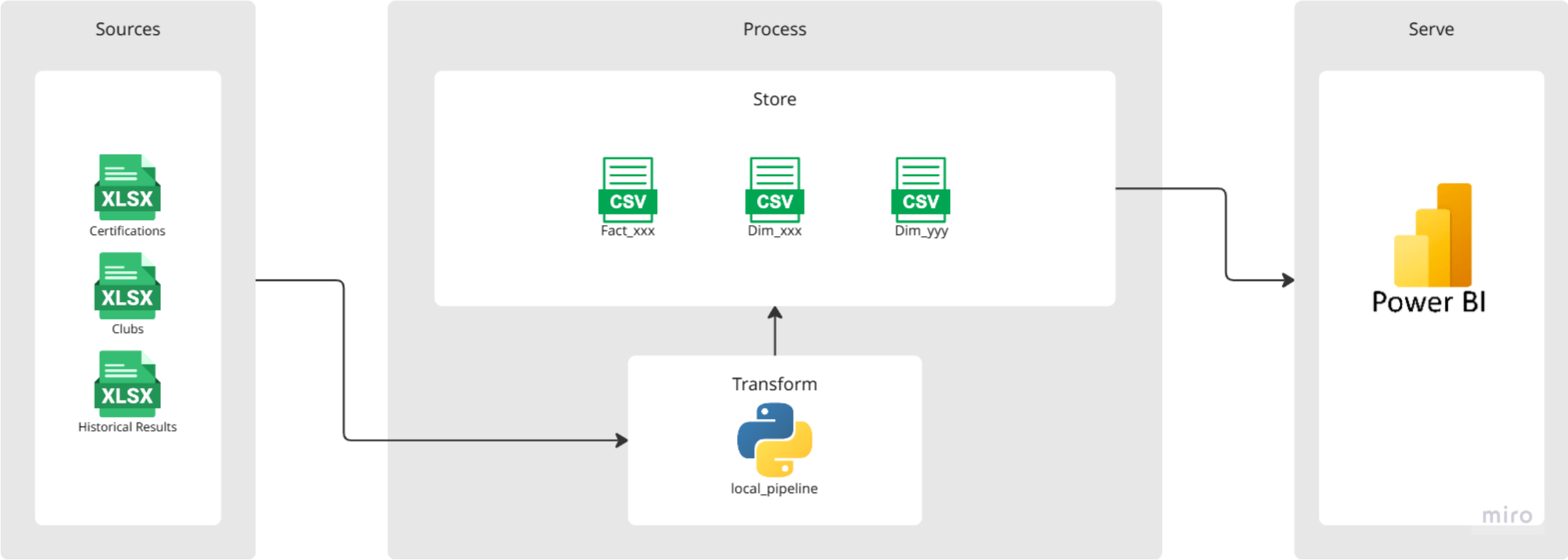
The Use Cases	The Data	The Client
• A list of **business questions** is available via the PDF on Canvas: • **Special Olympics - Use-Cases of Analytics.pdf**	• **Raw XLSX files** exported from the ERP system • Certifications, Clubs, and Historical Results. • **Special Olympics - Raw Data.zip** *(see Canvas)*	• **Bilal** (your lecturer) acts as the **primary stakeholder**. • Treat all **communication** and queries as you would with a **professional** client.

Primary Reference

- Before you begin, familiarize yourself with the official **documentation**.
- Search for "**Microsoft Power BI and Fabric Documentation**" on the official Microsoft Learn portal.

1. Resources

Base solution design:



2. Project Phases and Tasks

Phase 0: Project Management

Task:

- Create a detailed **weekly breakdown of your project** tasks.

Requirements:

- Address all **deadlines** correctly and ensure a logical **flow of development**.

Deliverable:

- **r0123456_ProjectPlan.pdf**
- Must be submitted as your first milestone.

2. Project Phases and Tasks

Phase 1: Dimensional Modeling

Task:

- Design a **Star Schema** including Fact and Dimension tables that **supports all use cases**.

Requirements:

- Ensure the **model** is **normalized** and addresses the provided PDF use cases.

Deliverable:

- **r0123456_Model.pdf**
- Exported from Lucidchart, Miro or a similar tool.

2. Project Phases and Tasks

Phase 2: The Python Data Pipeline (ETL) and Versioning

Task:

- Write a **Python script(s)** to **clean** the raw Excel files and **transform** them into the tables defined in your Star Schema.
- Each table is stored in an apart **CSV** file.

Requirements:

- Respect **OOP** principles.
- Use **GitHub** for version control.
- Your repository should include a **main.py** file, a **helpers folder** for functions, and a **data folder** for storing data.
- Raw (excel files) and clean(csv files) **data should be in different folders.**
- Use a **naming convention** e.g. dim_xxx.csv, fact_xxx.csv.

Deliverable:

- **r0123456_Repo_Archive.zip** (A zipped version of your full repository including the .git folder)
- **r0123456_Processed_Data.zip** (Folder containing your cleaned CSV files)
- ***r0123456_DatabaseExport.sql*** *(Only if you completed the MySQL bonus challenge; see the next page).*

2. Project Phases and Tasks

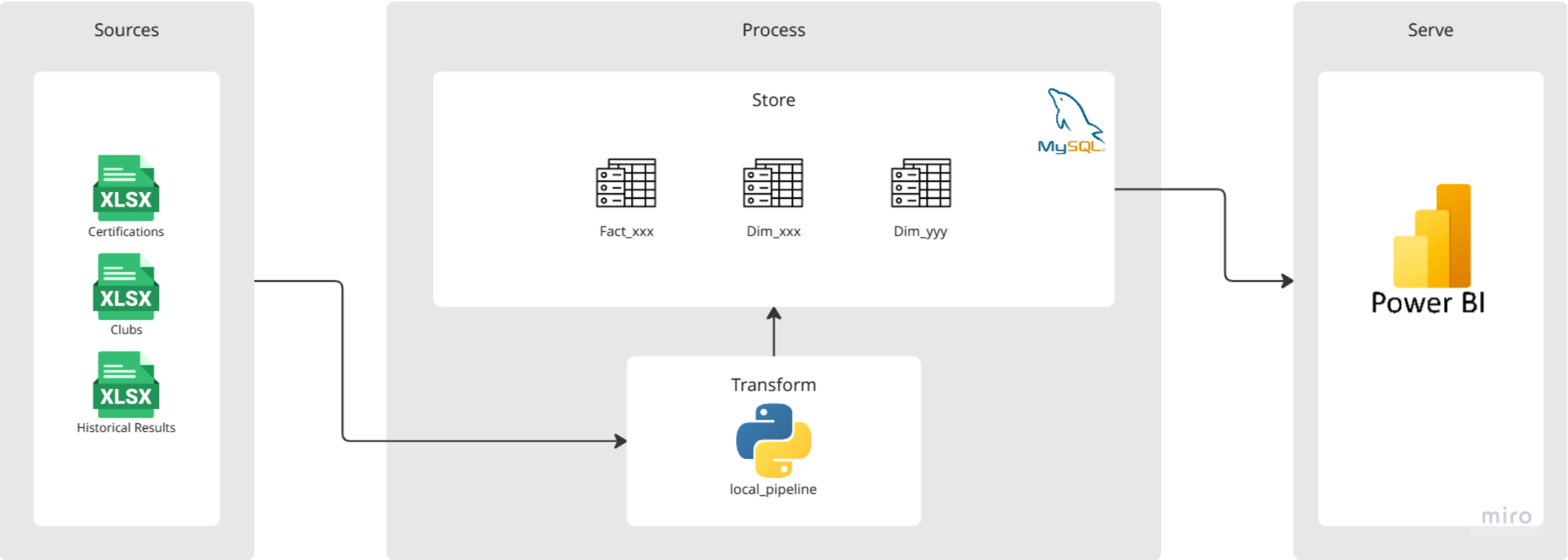
Phase 2: The Python Data Pipeline (ETL) and Versioning

Bonus Challenge 🏆:
You may modify your pipeline to **ingest the cleaned data into a MySQL Workbench** schema and **connect Power BI directly to that database**.

2. Project Phases and Tasks

Phase 2: The Python Data Pipeline (ETL) and Versioning

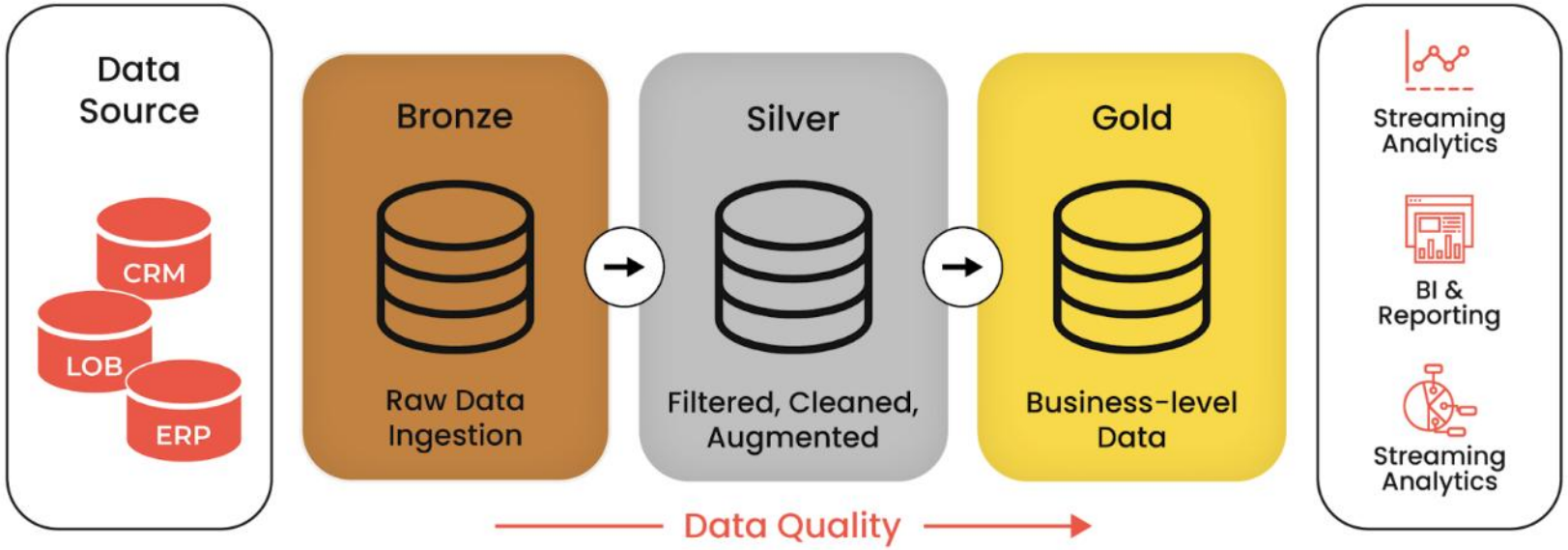
Base solution design + MySQL



2. Project Phases and Tasks

Phase 2: The Python Data Pipeline (ETL) and Versioning

Even More Bonus Challenge 🏆 🏆 :
You may use the recommended **medallion architecture** of **Microsoft**:



2. Project Phases and Tasks

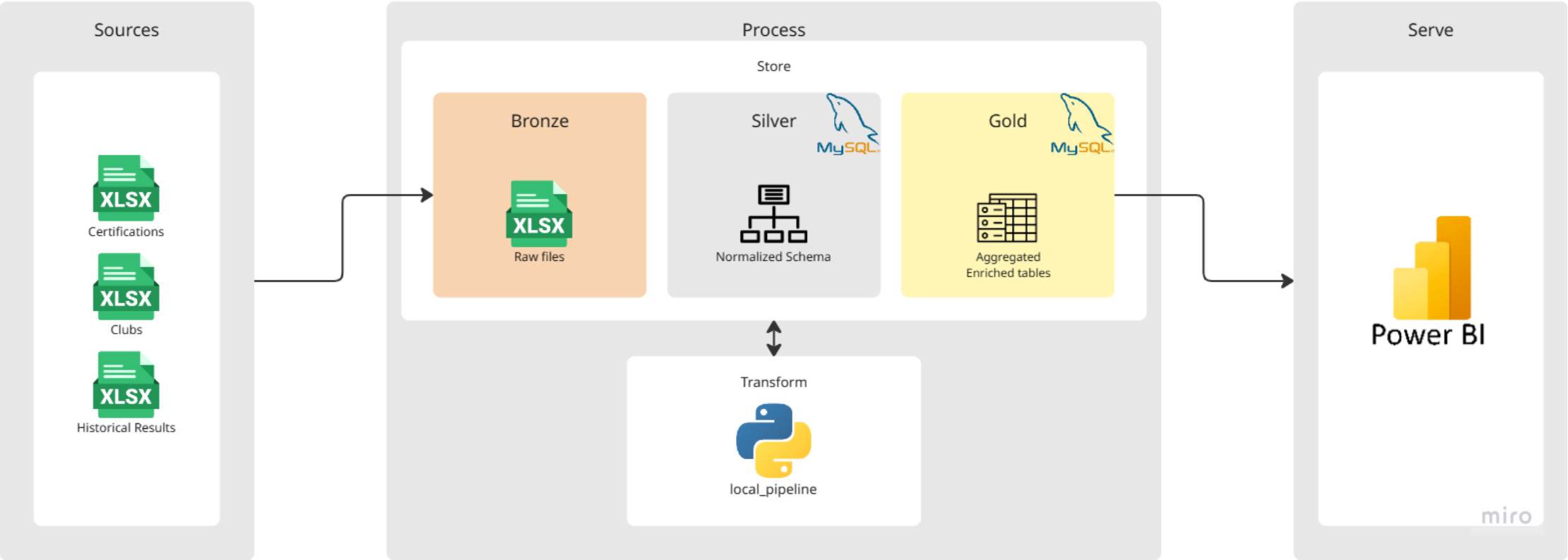
Phase 2: The Python Data Pipeline (ETL) and Versioning

- *Medallion architecture:*
  - *is **a data design pattern***
  - *used to **incrementally improve the structure and quality of data** as it flows through three distinct layers:*
    - ***Bronze (raw):***
      - ***Entry point** for data from various sources (IoT, databases, API). **Data** is ingested **in its raw format**, with full historical archives retained.*
    - ***Silver (cleansed):***
      - *Data from the Bronze layer is **cleaned**, **filtered**, **conformed**, and **joined**.*
    - ***Gold Layer (Enriched/Business Data):***
      - *The final layer represents data **organized** into actionable, **project-specific datasets**. It is structured for high-performance reporting, analytics, and machine learning.*

2. Project Phases and Tasks

Phase 2: The Python Data Pipeline (ETL) and Versioning

Base solution design + Medallion Arch. + MySQL



2. Project Phases and Tasks

Phase 3: Semantic Modeling and Power BI

Task:

- **Import** your **data** and establish the **relationships** between tables.

Requirements:

- All **heavy transformations** must happen **in the Python pipeline**.
- **Only minor adjustments** are allowed **in Power BI.**
- e.g. transforming **data types**, creating **measures** etc.

Deliverable:

- *Included in the final **r0123456_Dashboard.pbix** file.*

2. Project Phases and Tasks

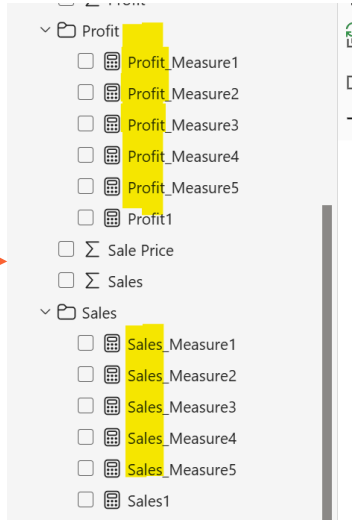
Phase 4: Design and Development

Task:

- Create a **wireframe** (mock-up report) for your report.
- Build a professional **multi page Power BI dashboard**.

Requirements:

- Design the dashboard logically;
  - visuals of similar use cases on the same **pages**.
  - High level info **vs** details.
- All **DAX measures** must be **organized** into dedicated folders.



Deliverable:

- **r0123456_Wireframe.pdf (or PNG).**
- **r0123456_Dashboard.pbix**

2. Project Phases and Tasks

Phase 5: Documentation and Validation

Task:

- Create a technical **documentation** file that **includes a data audit**.

Requirements:

- Include a "**Validation**" section with **screenshots showing** that your dashboard **totals match** the raw Excel source data.
- No need to validate every number on the dashboard; but you've got the idea!
- If you have questions for the client, **document** the **answers received** from the Q and A page here.

Deliverable:

- **r0123456_Documentation.pdf**

3. Naming Convention and Submission

This is an individual assignment, no group work allowed. *(you can share ideas, not files or work)*

Your **final submission** must be **a single ZIP file** named with your student number: **r0123456.zip**.
Inside this main ZIP, you must include:

1. r0123456_**ProjectPlan**.pdf
2. r0123456_**Model**.pdf
3. r0123456_**Wireframe**.pdf
4. r0123456_**Dashboard**.pbix
5. r0123456_**Documentation**.pdf
6. r0123456_**Processed_Data** (Folder containing your cleaned CSV files)
7. r0123456_**Repo_Archive**.zip
8. *r0123456_**DatabaseExport**.sql (If applicable)*

4. Important Rules

Q and A Page:

- Use the dedicated forum for any questions regarding the use cases.
- Your lecturer will act as the client to provide clarifications.

AI Policy:

- You **may use AI** but you must **document its usage**.
- **Never upload specific company or athlete data into AI tools.** ⚠️
- Keep your prompts generic.

Professionalism:

- This is a real-world example that will prepare you for your future career.
- Accuracy and intuitive design are your top priorities.

5. Evaluation

Category	Criterion	Excellent (Full Points)	Satisfactory (Partial)	Needs Improvement (Low)	Points
1. Management & Modeling (20)	Phase 0: Project Plan	Plan is highly realistic with logical weekly task breakdowns.	Plan exists but lacks detail or realistic timing.	Plan is missing or illogical.	5
	Phase 1: Dimensional Model	Star Schema is perfect. Facts and Dimensions are correctly identified and support all use cases.	Model is mostly correct but has minor logic flaws in relationships.	Incorrect model type or fails to address use cases.	15
2. Python Pipeline & Versioning (35)	ETL Logic & OOP	Code is clean, follows OOP principles, and handles all logic in Python efficiently.	Logic is correct but OOP is weak or code is messy.	Heavy use of manual steps or inefficient transformations.	15
	Versioning (GitHub)	Professional workflow with frequent, descriptive commits showing clear progress.	Commits exist but are irregular or have vague messages.	Only one or two "final" commits; no version history.	10
	Code Structure	Follows best practices with main.py, helpersfolder, and clear naming.	Structure is mostly organized but lacks standard folder naming.	Messy file structure; difficult to navigate code.	10
3. Power BI Development (25)	Wireframe	Clear, professional UI/UX plan that precedes development.	Basic wireframe provided but lacks detail.	No wireframe or very poor design plan.	5
	Semantic Model & Measures	Correct relationships; DAX measures are perfectly organized in folders.	Relationships work but measures are unorganized.	Incorrect connections or broken DAX measures.	10
	Actionable Dashboard	Professional, easy to read, and provides direct answers to client questions.	Good visuals but answers are hard to find or interpret.	Poor design; does not answer the business questions.	10
4. Documentation & Validation (20)	Data Dictionary	Every column and calculation is clearly defined for a non technical user.	Most columns defined but some definitions are vague.	Missing or incomplete dictionary.	10
	Validation Audit	Screenshots provide absolute proof that numbers match raw source files.	Validation is attempted but lacks clear proof or screenshots.	No validation provided; numbers cannot be trusted.	10
TOTAL					100

5. Evaluation

Bonus points:

Category	Criterion	Requirement	Points
Advanced Stack	**MySQL Integration**	Pipeline ingests data into MySQL and connects to Power BI. Must	**10**
	Medallion Architecture	Data is clearly layered as the architecture suggests.	**10**

5. Evaluation

Important Notes for Grading

- **Naming Convention:**
  - Deduct 5 points if the naming format is ignored in any part of the project.
- **AI Policy:**
  - Zero grade if undocumented AI usage is identified.
- **Late Submissions:**
  - As per school policy, 0 points for the missing/late submissions unless a doctor's note or formal excuse is provided.

6. Pro Tips for Success

- This project takes time!
- **Cannot be finished in the final weeks!**

Start Now

- **Search** for **best practices**.
- It is **your daily job**. If you are stuck, research it.

Researcher Mindset

- You are **100% responsible** for validating everything.

AI is the **Assistant, not the Pilot**

- It is much **harder** to document **at the end** if you didn't capture the process.

Document while Building

“The best way out is always through.”

Robert Frost